

Yuchen (Jency) Jiang

+1 (814) 280 8680 | yuchen_jiang2@brown.edu

EDUCATION

- Aug 2024 – Present** Doctorate in Applied Mathematics, Brown University
- Pursuing a secondary master in cognitive sciences and psychology supported by Open Graduate Fellowship
- Aug 2023 – May 2024** Doctorate in Mathematics, Pennsylvania State University
- Transferred into Brown University
- July 2021 – Dec 2022** Master of Mathematical Sciences (Advanced), Australian National University
- GPA: 6.9/7.0 with First Class Thesis
- July 2018 – July 2021** Bachelor of Mathematical Sciences (Advanced), University of Adelaide
- GPA: 7.0/7.0, major in pure mathematics and statistics

RESEARCH INTERESTS

I am interested in applying topological, combinatorial and algebraic methods to understand neural dynamics and, more generally, developing novel tools to tackle questions in cognitive sciences and neuroscience.

PROGRAMMING LANGUAGES

Proficient in Matlab, Python, and R.

PUBLICATION

1. **Y. J. Jiang**, V. Robins and K. Turner 2026, *Computing extended persistent homology of radial distance filtrations of Euclidean shapes*, <http://arxiv.org/abs/2608.11963>.

AWARDS & SCHOLARSHIPS

2026 –	Open Graduate Education Fellowship <i>Awarded to high-achieving graduate students at Brown to pursue a secondary master's degree</i>
2023 – 2024	Maryam Mirzakhani Science Achievement Graduate Fellowship in Mathematics <i>Awarded at PSU for one female student in Mathematics each year demonstrating leadership skills</i>
2024 Spring	George E. Bergey Mathematics Discretionary Endowment at PSU
2024 Spring	University Graduate Fellowship <i>Prestigious scholarship at PSU for incoming graduate students</i>
2023 Fall	Verne M. Willaman Distinguished Graduate Fellowship in Science at PSU
2023 Fall	Jack and Eleanor Pettit Scholarship in Science at PSU
2021, 2022	ANU Chancellor's International Scholarship <i>Awarded at ANU for international master students based on academic merit</i>
2021 Summer	Meritorious Winner in Mathematical Contest in Modeling <i>Participated in a team of three</i>
2019, 2020	The Executive Dean's Recognition of Academic Excellence <i>Awarded at UoA for a year-long high-distinction GPA (i.e. 7.0/7.0)</i>
2019 – 2020	AMSI Vacation Research Scholarship <i>Summer research program hosted at the University of Melbourne (Project: Exploring Toposes)</i>

TEACHING AND LEADERSHIP EXPERIENCE

- Summer 2026 **Instructor**, Brown
- MATH0520 (Linear Algebra)

- 2025 – 2026 **Graduate Teaching Assistance**, Brown
- *APMA 1210 (Operations Research: Deterministic Models)*
 - *APMA 1200 (Operations Research: Probabilistic Models)*

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- 2021 – 2022 **Demonstrator**, ANU
- *This position is equivalent to a TA.*
 - *MATHS 1115 (Advanced Mathematics & Applications I)*
 - *MATHS 1116 (Advanced Mathematics & Applications II)*
 - *Awarded College of Science Dean's Commendation for Excellence In Education 2023*

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- 2020 – 2023 **Private Math Tutor for Tertiary Students**, Online
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WORKSHOPS, CONFERENCES & SUMMER SCHOOLS

- July 2026 **Mathematical Methods in Computational Neuroscience**,
Fred Kavli Knowledge Center, Eresfjord, Norway

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- February 2026 **Neuroscience and ML Workshop**
University of Chicago Neuroscience Institute and Data Science Institute

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- November 2025 **Junior Scientist Workshop on Theoretical Neuroscience**,
HMMI-Janelia Research Campus

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- Spring 2025 **AATRN Tutorial-a-thon**, Virtual
- *Recorded a short tutorial on extended persistent homology (can be found [here](#)).*

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- July 2024 **Theoretical and Computational Neuroscience Summer School (CNeuro2024)**,
Tsinghua University and IOB Basel

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- October 2023 **Topology and Geometry in Neuroscience**, ICERM
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- September 2022 **AATRN Poster Session**, Virtual
- *Presented a poster based on work for my master's thesis in applied topology.*